



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

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|---------------|
| JEE(Advanced) |
| REVIEW TEST   |
| 21-08-2021    |

JEE(Main + Advanced) : NURTURE COURSE PHASE-TEPS, TOPS, I, I(A), I(B), II(A) & NURTURE LIVE -I, I(B), II(A)

Time : 3 Hours

## PAPER-2

Maximum Marks : 198

### READ THE INSTRUCTIONS CAREFULLY

#### GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

#### OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

#### DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take  **$g = 10 \text{ m/s}^2$**  unless otherwise stated.

#### QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

**SOME USEFUL CONSTANTS**

**Atomic No.** : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,  
Cl = 17, Br = 35, Xe = 54, Ce = 58

**Atomic masses :** H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,  
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,  
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

**Space for Rough Work**

**HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS**

**BEWARE OF NEGATIVE MARKING**

**PART-1 : PHYSICS**

**SECTION-I : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

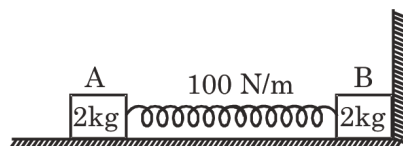
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

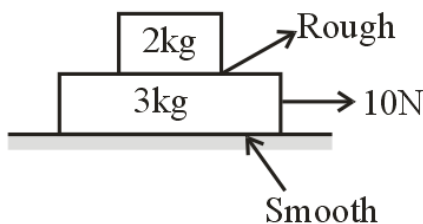
1. Initially spring is compressed by 0.5 m and surface is smooth. Choose the correct option(s) :-



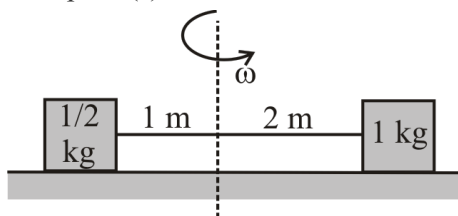
- (A) Initial potential energy in spring is 12.5 J.
- (B) Kinetic energy of block A, when spring is at natural length for first time is 12.5 J.
- (C) Work done by all forces on B is zero till spring attains it's natural length for first time.
- (D) None of these.

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2. In the figure shown ground is smooth, whereas surface between 3 kg block and 2 kg block is rough. A force of magnitude 10 N is applied on 3 kg block, horizontally. If net work done by friction on the system (on 3 kg and 2 kg blocks) in any time frame is zero, then coefficient of friction between 3 kg and 2 kg can be :



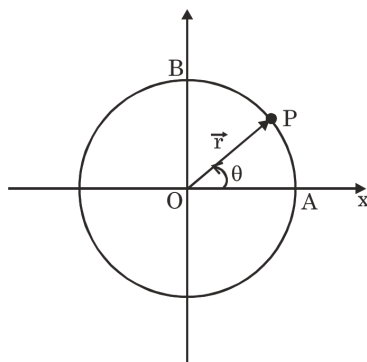
- (A) 0.1                      (B) 0.3                      (C) 0.4                      (D) 0.6
3. Two small blocks tied with a massless string of length 3 m are placed on a rotating table as shown. The axis of rotation is 1 m from  $\frac{1}{2}$  kg mass and 2 m from 1 kg mass. The angular speed  $\omega = 1$  rad/s and the blocks do not slip. Surface below 1 kg block is smooth and that below  $\frac{1}{2}$  kg block is rough (friction coefficient = 0.6). Choose the correct option(s).



- (A) Tension in the string is 2 N                      (B) Tension in the string is 1 N
- (C) Friction force on  $\frac{1}{2}$  kg block is  $\frac{3}{2}$  N                      (D) Friction force on  $\frac{1}{2}$  kg block is  $\frac{5}{2}$  N

Space for Rough Work

4. A particle P performs circular motion such that its position vector  $\vec{r}$  makes an angle  $\theta$  with x axis according to following equation  $\theta = t^2 - 6t + 8$  rad.

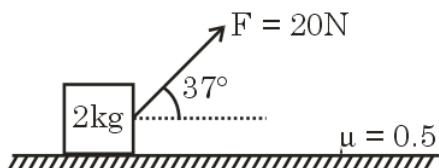


- (A) At  $t = 0$  particle is at point A.  
 (B) Particle changes its direction of motion only once.  
 (C) Particle does not change it's direction of motion.  
 (D) Angular acceleration of particle is constant.

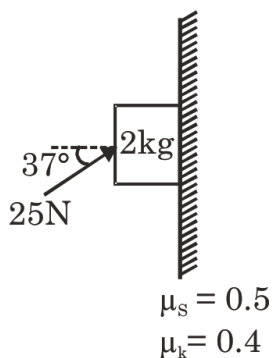
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Space for Rough Work

5. A block of mass 2 kg initially is at rest on a rough surface having coefficient of friction  $\mu = 0.5$ . A force 20N is applied as shown in figure then which of the following is/are **CORRECT** :



- (A) Acceleration of block is  $6\text{m/s}^2$ .  
 (B) Work done by applied force (F) when block has moved by 2m is 32J.  
 (C) Work done by kinetic friction when block has moved by 2m is -8J  
 (D) Magnitude of normal contact force applied by block on the ground is 8N.
6. 25 N force is applied to the block as shown in figure. Choose the correct option(s) if initial velocity of block is zero :-



- (A) Normal contact force is 20N.  
 (B) Friction force is 5N.  
 (C) Friction force is 10N.  
 (D) Block is in equilibrium.

**Space for Rough Work**

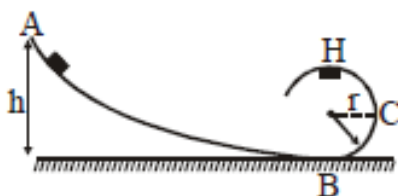
**SECTION-II : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
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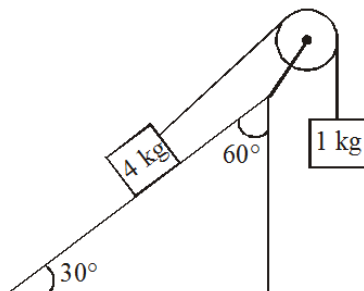
*Zero Marks* : 0 In all other cases.

1. A spring balance is attached to the ceiling of a lift. A man hangs his bag on the spring and the spring reads 49 N, when the lift is stationary. If the lift moves downward with an acceleration of  $5 \text{ m/s}^2$ , the reading of the spring balance (in N) will be ( $g = 9.8 \text{ m/s}^2$ )-
2. A small block of mass  $m$  is allowed to slide from  $h = 2r$  on an inclined frictionless track in vertical plane from rest position as shown in the figure. Find total acceleration of block at point C is



Space for Rough Work

3. A particle moves with a velocity  $\vec{v} = (5\hat{i} - 3\hat{j} + 6\hat{k})\text{m/s}$  under the influence of a constant force  $\vec{F} = (10\hat{i} + 10\hat{j} + 20\hat{k})\text{N}$ . The instantaneous power applied to the particle is (in J/s) :
4. A man wishes to cross a river of width 120 m by a motorboat. His rowing speed in still water is 5 m/s. He takes  $t$  seconds to cross the river and reach to the point directly opposite to his starting point. Speed of water is 3 m/s. Find  $t$ .
5. A piece of marble is projected from earth's surface with velocity of 25 m/s. 2 seconds later, it just clears a wall 5 m high. What is the angle of projection (in degree) ?
6. Find the tension (In N) in the cord in the device. Assume no friction anywhere.




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Space for Rough Work



**SECTION-III : (Maximum Marks: 18)**

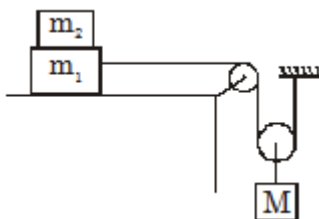
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- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
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*Zero Marks* : 0 If no answer is given.

*Negative Marks* : -1 In all other cases

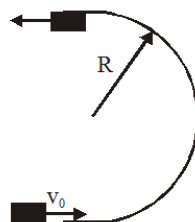
1. Find  $M$  (in kg) for which  $m_2$  will start sliding over  $m_1$ . Table on which  $m_1$  and  $m_2$  are placed is smooth,  $\mu$  is the friction coefficient between  $m_1$  and  $m_2$ . (Given :  $m_1 = 2$  kg,  $m_2 = 1$  kg,  $\mu = 0.5$ ,  $g = 10$  m/s<sup>2</sup>)



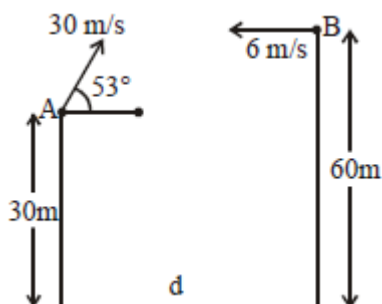
2. A single conservative force  $F(x)$  acts on 1 kg particle that moves along  $x$ -axis. The potential energy  $U(x)$  is given by  $U(x) = (x-2)^2 - 20$ . Where  $x$  is in meters and  $U(x)$  in joules. If particle has a kinetic energy of 20 J at  $x = 5$  m, find mechanical energy (in joules) at  $x = 2$  m.

**Space for Rough Work**

3. Figure shows a bar of mass 'm' moving with a constant speed  $v_0$  on horizontal surface around a semi-circular fence of radius R. The average normal force acting on the bar due to contact with the fence is given by  $\sqrt{\alpha} \frac{mv_0^2}{\pi R}$ . Fill the value of  $\alpha$ . Consider all surfaces to be smooth.

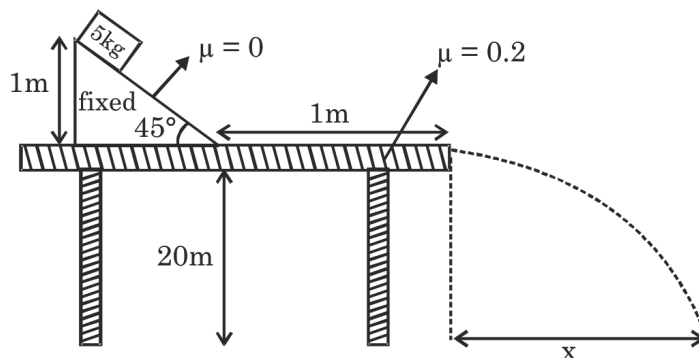


4. Ball A is projected from tower of height 30 m and B is projected horizontally at 6 m/s. A and B collide in air. If separation between the towers is  $d = 10x$ . What is the value of x.



Space for Rough Work

5. A certain system has potential energy given by the function  $U(x) = -8x^2 + x^4$ . What is the x coordinate of the point where the equilibrium is unstable ?
6. A block of mass = 5kg released from rest on the top of the smooth inclined plane which is fixed to table as shown in figure. It then slides on a rough horizontal surface of  $\mu = 0.2$  for 1m. It then leaves the top of the table which is 20.0 m high. How far from the base of the table does the block lands.



Space for Rough Work

**PART-2 : CHEMISTRY**  
**SECTION-I : (Maximum Marks: 24)**

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1. Successive ionisation energies for some elements are listed below, from which predict that the elements are of same group in the following options.

|     | IE <sub>1</sub> | IE <sub>2</sub> | IE <sub>3</sub> | IE <sub>4</sub> | IE <sub>5</sub> | IE <sub>6</sub> | IE <sub>7</sub> |
|-----|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| J : | 14.534          | 29.601          | 47.448          | 77.472          | 97.888          | 552.057         | 667.029         |
| K : | 8.151           | 16.345          | 33.492          | 45.141          | 166.77          | 205.05          | 246.52          |
| L : | 17.422          | 34.970          | 62.707          | 87.138          | 114.240         | 157.161         | 185.182         |
| M : | 10.486          | 19.725          | 30.18           | 51.37           | 65.023          | 220.43          | 263.22          |
| N : | 11.26           | 24.383          | 47.887          | 64.492          | 392.077         | 489.981         |                 |

(A) J and K

(B) K and N

(C) L and M

(D) J and M

Space for Rough Work

2. Which of the following species is/are planar, polar and hypervalent?  
(A)  $\text{XeF}_5^\ominus$  (B)  $\text{SO}_2$  (C)  $\text{XeO}_6^{4-}$  (D)  $\text{ClF}_3$
3. A solution of  $\text{C}_2\text{H}_5\text{OH}$  & water contains 54% water by mass, then which option(s) is/are correct for the given solution- [Given density of solution = 1 gm/ml]  
(A) Molality = 18.52 (B) % w/w = 46 (C)  $X_{\text{C}_2\text{H}_5\text{OH}} = 0.25$  (D) % w/v = 46
4. Choose the incorrect statement(s) :  
(A) The oxidation number of iodine in  $\text{HIO}_4$  is + 6  
(B) In the reaction  $\text{Zn} + \text{Cu}(\text{NO}_3)_2 \rightarrow \text{Cu} + \text{Zn}(\text{NO}_3)_2$ , the Zn is reduced  
(C) In the reaction  $\text{Cu}_2\text{S} + \text{O}_2 \rightarrow \text{CuO} + \text{SO}_2$ , the n-factor for  $\text{Cu}_2\text{S}$  is 8.  
(D) All 'S' atoms are present in same oxidation state in sodium thiosulphate(hypo).

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**Space for Rough Work**

5. 34 gm of  $\text{H}_2\text{O}_2$  dissolved in water to make a solution of 2 litre, then choose the correct options.
- (A) Volume of strength of  $\text{H}_2\text{O}_2$  is 5.675 V  $\text{H}_2\text{O}_2$
- (B) Molality of this solution is 2 m
- (C) % w/v of this solution is 1.7% w/v
- (D) Molarity of  $\text{H}_2\text{O}_2 = 1\text{M}$
6. Which among the following pair of molecules have same shape and same hybridisation state :-
- (A)  $\text{XeO}_2\text{F}_2$ ,  $\text{SF}_4$       (B)  $\text{PCl}_3$ ,  $\text{BF}_3$       (C)  $\text{I}_3^+$ ,  $\text{XeF}_2$       (D)  $\text{ClO}_4^-$ ,  $\text{CH}_4$
- 

Space for Rough Work

**SECTION-II : (Maximum Marks: 24)**

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- 
1. How many maximum number of electrons a subshell with  $n = 6$ ,  $\ell = 2$  can accommodate ?
  2. The formation of  $O_{(g)}^{2-}$  starting from  $O_{(g)}$  is endothermic by  $639 \text{ kJ mol}^{-1}$ . If electron gain enthalpy of  $O_{(g)}$  is  $-141 \text{ kJ mol}^{-1}$ , the second electron gain enthalpy(KJ/mol) of oxygen would be :
  3. 62.5 gm of a mixture of  $CaCO_3$  and  $SiO_2$  are treated with excess of  $HCl$  due to which 1.1 gm of  $CO_2$  is produced. What is the mass % of  $CaCO_3$  in the mixture?
  4. Calculate molarity of 49% w/w  $H_2SO_4(aq.)$  if density of solution is 1.8 g/ml.
  5. In an organic compound of molar mass  $108 \text{ g mol}^{-1}$  C, H and N atoms are present in 9 : 1: 3.5 by weight. Molecular formula of compound is  $C_xH_yO_z$ . The value of  $\frac{x+y+z}{5}$  is-
  6. Calculate the number of  $\sigma$  bond in given compound  $C_3O_2$
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**Space for Rough Work**

**SECTION-III : (Maximum Marks: 18)**

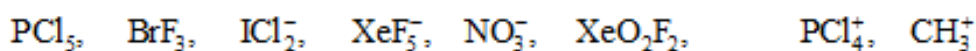
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1. Find out the number of pairs in which boiling point of first compound is less than second?  
(B(Me)<sub>3</sub>, BF<sub>3</sub>) ; (SiCl<sub>4</sub>, CCl<sub>4</sub>) ; (H<sub>2</sub>, O<sub>2</sub>) ; (NF<sub>3</sub>, N(Me)<sub>3</sub>) ; (H<sub>3</sub>BO<sub>3</sub>, B(OMe)<sub>3</sub>) ; (CO, N<sub>2</sub>) ; (H<sub>2</sub>O, HF)
2. For the following molecules :



Calculate the value of  $\frac{a+b}{c}$

a = Number of species having sp<sup>3</sup>d-hybridisation

b = Number of species which are planar

c = Number of species which are non-planar

**Space for Rough Work**



3. For the given redox reaction,  $x\text{KMnO}_4 + y\text{H}_2\text{O}_2 + z\text{H}_2\text{SO}_4 \rightarrow p\text{MnSO}_4 + 10\text{O}_{2(g)} + q\text{K}_2\text{SO}_4 + r\text{H}_2\text{O}$ .  
Calculate value of  $(x + z - p)$ .
4. In order to oxidise 5 moles of  $\text{Fe}_2(\text{C}_2\text{O}_4)_3$  in acidic medium, then number of moles of  $\text{KMnO}_4$  required will be :
5. Based on following reaction, give minimum possible value of 'x' for 1 mol  $\text{AF}_6$ .  
$$\text{AF}_6 + \text{H}_2\text{O} \rightarrow \text{AO}_x + \text{HF}$$
6. Find the number of ions having ionic radius greater than  $\text{Na}^+$ .  
 $\text{K}^+, \text{Rb}^+, \text{Mg}^{+2}, \text{F}^-, \text{O}^{2-}, \text{Al}^{+3}$
- 

**Space for Rough Work**

**PART-3 : MATHEMATICS**  
**SECTION-I : (Maximum Marks: 24)**

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1. In a  $\Delta ABC$  if  $2\sin A \cos C = \sin B$  then which of the following is/are true

(A)  $\frac{a}{\cos A} = \frac{c}{\cos C}$

(B)  $a^2 + c^2 = ac$

(C) length of median through vertices A and C are equal

(D) incentre, orthocentre, centroid and circumcentre are collinear

**Space for Rough Work**

2. In  $\triangle ABC$ , with usual notation, if a semicircle is drawn having its diameter on side AB and touching the sides AC and BC, then radius of semicircle can be expressed as -

(A)  $\frac{2abc}{s(a+b)} \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$

(B)  $\frac{2abc}{s(a+b)} \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$

(C)  $\frac{a\Delta}{2bc}$

(D)  $\frac{2\Delta}{a+b}$

3. If  $T_r = \begin{vmatrix} r-1 & r-2 & r-6 \\ 2r-4 & 2r-6 & 2r-11 \\ 3r-9 & 3r-12 & 3r-18 \end{vmatrix}$ , then -

(A)  $\sum_{r=0}^5 T_r = 36$

(B)  $\sum_{r=1}^5 T_r = 30$

(C)  $\prod_{r=0}^5 T_r = 6^6$

(D)  $\prod_{r=1}^5 T_r = 6^5$

4. The number of ordered pairs (x,y) satisfying  $|x| + |y| = 3$  and  $\sin\left(\frac{\pi x^2}{3}\right) = 1$  is less than or equal to

(A) 10

(B) 9

(C) 8

(D) 7

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Space for Rough Work

5. If solution of the equation  $3\sin x + 4\cos x = \max\{1, 6 + a^2 + 2a; a \in \mathbb{R}\}$  exist then
- (A) number of values of  $x$  satisfying above equation in  $[0, 2\pi]$  is 2  
(B) number of values of  $x$  satisfying above equation in  $[0, 2\pi]$  is 1  
(C)  $a = 1$   
(D)  $a = -1$
6. If the  $n^{\text{th}}$  term of an arithmetic progression is  $T_n = (a - 4)n^3 + (b - 7)n^2 + (a + b)n - 8 \forall n \in \mathbb{N}$ , (where  $a, b \in \mathbb{R}$ ) then which of the following is/are true-
- (A)  $a^3 - b^2 = 15$  (B) common difference of the AP is 11  
(C) common difference of the A.P is 13 (D) sum of first 8 terms of the A.P is 332
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**SECTION-II : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If ONLY the correct numerical value is entered.

*Zero Marks* : 0 In all other cases.

1. If ABC is a triangle such that its incentre 'I' is equidistant from its vertices A, B and C and radius of its incircle is 4, then the area of the triangle ABC (Take  $\sqrt{3} = 1.7$ ) is :-
2. Let  $f(\theta) = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 + \sin \theta & 1 \\ 1 & 1 & 1 + \cos \theta \end{vmatrix}$ . Number of solutions of the equation  $f(\theta) = 0$  in  $\theta \in (0, 2\pi)$  is
3. If  $2 + 22 + 222 + 2222 + \dots$  10 term  $= \frac{a(10^{11} - 10^a)}{c^a}$  (Where  $a, c \in \mathbb{N}$ ) then value of  $\left(\frac{c}{a}\right)$  is
4. If the roots of the equation  $x^2 - bx + c = 0$  be two consecutive integers, then  $(b^2 - 4c)$  equals-
5. In a  $\Delta ABC$  (with usual notations) let  $a : b : c = 4 : 5 : 6$  then  $\frac{8r}{7R}$  is
6. Number of value(s) of  $x \in [0, 2\pi]$  which satisfies the equation  $\sin^2 x + \operatorname{cosec}^2 x = \cos^2 x + \sec^2 x$  is

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SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +3 If only the correct answer is given.

*Zero Marks* : 0 If no answer is given.

*Negative Marks* : -1 In all other cases

1. In  $\triangle ABC$  if angles A, B, C (In order) are in A. P. &  $\angle A$  exceeds lowest angle by  $30^\circ$  & D divides BC internally in 1:3 and  $\frac{\sin \angle BAD}{\sin \angle CAD} = \frac{1}{\sqrt{k}}$ . (Where  $k \in \mathbb{N}$ ) then the value of k is
2. Let  $f(x) = 25^{(-\cos x + 3^{-a})}$  where  $a = \log_3 \log_4 5$ . If  $80 < f(x) < 400$ , then sum of all integral value(s) of x in  $x \in [0, 2\pi]$ , is (take  $\pi = 3.14$ )
3. If the angles of a triangle are in A.P. with common difference equal to the least angle, so that the sides are in ratio  $a : \sqrt{b} : c$ , (where a, b, c are coprime pairwise), then the value of  $(a + b + c)$  is
4. In right angled triangle ABC, the value of  $\sin^2 A + \sin^2 B + \sin^2 C$  is  $\frac{k}{3}$ , then k is
5. Sum of all real solution of the equation  $|x - 3|^2 - |x - 3| - 2 = 0$  is
6. If  $x = \log_{2b} b$ ,  $y = \log_{3b} 2b$  and  $z = \log_{4b} 3b$  (wherever defined), then  $\frac{(1 + xyz)}{yz}$  is

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NAME OF THE CANDIDATE .....

FORM NO. ....

I have read all the instructions  
and shall abide by them.

\_\_\_\_\_  
Signature of the Candidate

I have verified the identity, name and Form  
number of the candidate, and that question  
paper and ORS codes are the same.

\_\_\_\_\_  
Signature of the Invigilator

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